

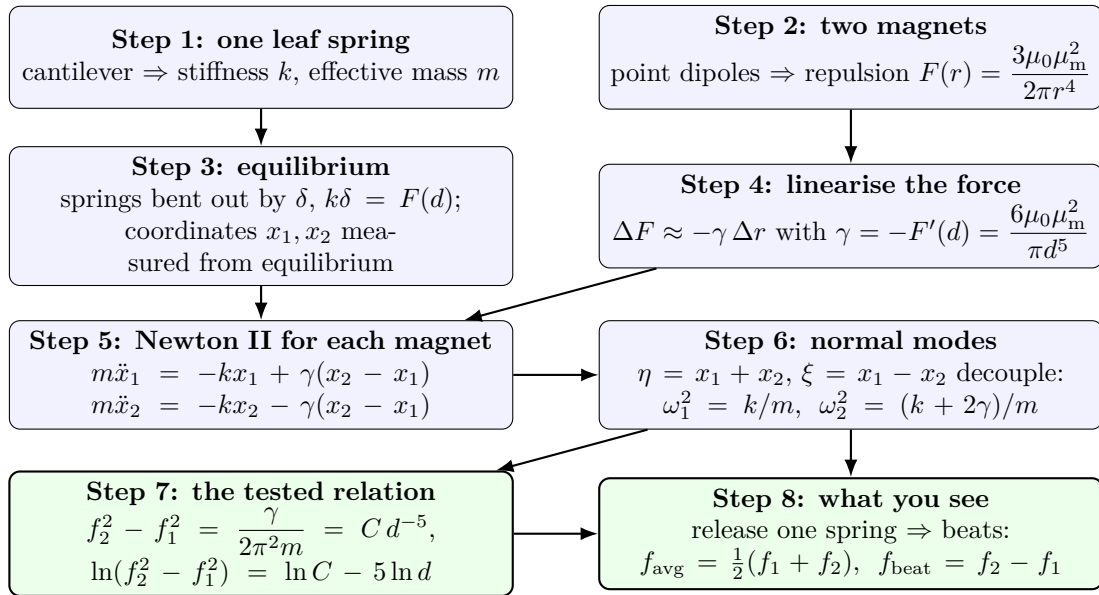
The theory of the coupled magnetic–mechanical oscillator

Section 5 of the Extended Essay, explained step by step

Study notes to accompany the essay draft

These notes go through the theoretical model of the essay (Section 5) one logical step at a time, with a diagram for every step. Each step has the same layout: the *idea* in one sentence, the *maths*, the *result*, and a box saying what you need to convey when you write it in your own words. The maths is exactly the same as in the essay; only the explanations are longer. Read the flow chart first: every later section is one box of it.

The whole argument on one page



The logic is: *mechanics* gives k and m ; *magnetism* gives $F(r)$; *linearising* F turns the magnetic repulsion into a third spring of stiffness γ ; two masses joined by three springs have two normal modes; the difference of the squared mode frequencies depends only on γ/m , and $\gamma \propto d^{-5}$.

1 Step 1: a leaf spring with a magnet is a mass on a spring

Idea in one sentence

A clamped strip pushed sideways at its free end pushes back with a force proportional to the deflection (Hooke's law), so the tip behaves like a spring of stiffness k ; the magnet plus part of the strip is the mass m on that spring.

1a. The stiffness k

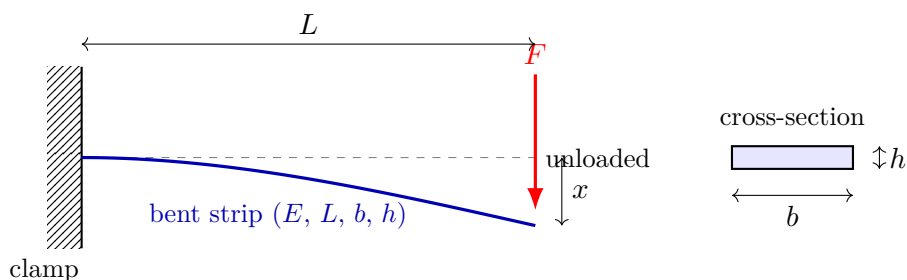


Figure 1. A cantilever: a strip of length L , width b , thickness h , clamped at one end and pushed sideways at the other. The tip moves by x .

Engineering beam theory (the standard cantilever formula, derived in Appendix A of the essay) says that for a tip force F

$$x = \frac{FL^3}{3EI}, \quad I = \frac{bh^3}{12}. \quad (1)$$

Here E is Young's modulus of the material (how stiff copper is) and I is the *second moment of area* of the cross-section, a purely geometrical number that says how well the shape resists bending. Rearranging, force divided by deflection is a constant, i.e. a spring constant:

$$k = \frac{F}{x} = \frac{3EI}{L^3} = \frac{Ebh^3}{4L^3} \quad (2)$$

Two sanity checks that show the formula is sensible: doubling the length makes the spring $2^3 = 8$ times *softer* (a long strip is floppy), doubling the thickness makes it 8 times *stiffer*, and doubling the width only doubles k (two strips side by side).

1b. The mass m

If the strip had no mass, the oscillating mass would just be the magnet, M . But the strip does have mass, and it moves too. The problem is that it does not all move equally: the part next to the clamp barely moves, the tip moves as much as the magnet (Figure 2).

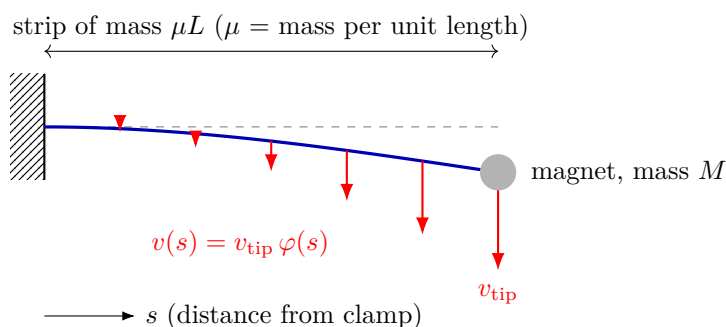


Figure 2. Rayleigh’s idea. Every point of the strip moves with a velocity proportional to the static bending shape $\varphi(s)$ ($\varphi = 0$ at the clamp, $\varphi = 1$ at the tip). The strip’s kinetic energy is the same as that of a point mass $\frac{33}{140}\mu L$ moving with the tip.

Rayleigh’s method turns this into one number. Assume the strip vibrates in the same shape as its static deflection curve, $\varphi(s) = \frac{s^2(3L-s)}{2L^3}$, so that a point at distance s from the clamp moves with velocity $v_{\text{tip}}\varphi(s)$. Its kinetic energy is

$$K_{\text{strip}} = \int_0^L \frac{1}{2} \mu (v_{\text{tip}}\varphi(s))^2 ds = \frac{1}{2} \left(\mu \int_0^L \varphi^2 ds \right) v_{\text{tip}}^2 = \frac{1}{2} \left(\frac{33}{140} \mu L \right) v_{\text{tip}}^2. \quad (3)$$

So, as far as kinetic energy is concerned, the strip behaves like a point mass $\frac{33}{140}\mu L \approx 0.24\mu L$ sitting at the tip (the integral $\int_0^L \varphi^2 ds = \frac{33}{140}L$ is just algebra with the cubic). The effective mass is therefore

$$m = M + \frac{33}{140} \mu L \quad (4)$$

and a single spring on its own oscillates at $\omega_0 = \sqrt{k/m}$, $f_0 = \omega_0/2\pi$.

Common confusion

m is *not* the mass of the magnet, and *not* the magnet plus the whole strip; it is the magnet plus about a quarter of the strip. If the magnet is much heavier than the strip the correction hardly matters, which is why heavy magnets make the lumped model accurate.

1c. Why we measure k and m instead of calculating them

The formulas contain E (uncertain by $\sim 10\%$ for rolled copper) and h^3 (a 3% error in thickness becomes 9% in k). So the formulas are used to *understand* the system, while the numbers are *measured*: plucking a spring with its own magnet (other magnet removed) gives $f_0 = \frac{1}{2\pi}\sqrt{k/m}$ directly, and a load–deflection graph gives k ; together they give m . For the exponent test (gradient –5) neither k nor m is needed at all, as Step 7 shows.

What to say in your own words

(1) A clamped leaf spring obeys Hooke’s law at its tip, with $k = 3EI/L^3$. (2) The oscillating mass is the magnet plus an effective fraction (33/140) of the strip, because the strip moves less near the clamp. (3) One spring alone is therefore a simple harmonic oscillator with $\omega_0^2 = k/m$. (4) k and m are measured rather than calculated because E and h are poorly known.

2 Step 2: the force between the two magnets

Idea in one sentence

Seen from a few diameters away, a small magnet is a magnetic dipole; the field of a dipole falls as $1/r^3$, the force on a second dipole is proportional to the *gradient* of that field, so the force between two facing magnets falls as $1/r^4$.

2a. The field of one magnet on its axis

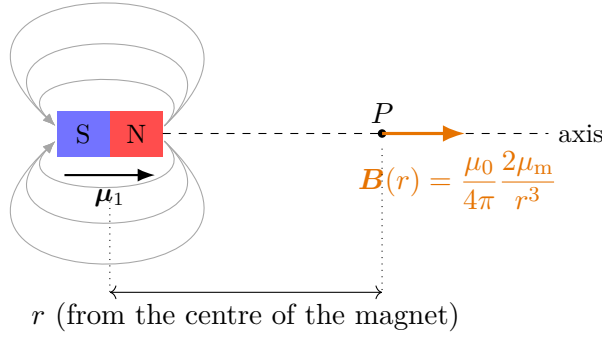


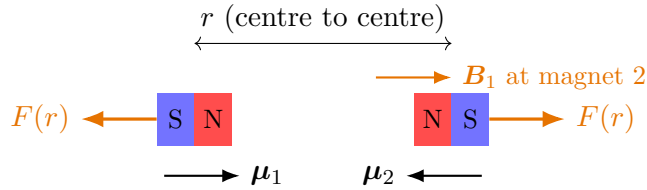
Figure 3. A small magnet is a dipole of moment μ_m . On its axis, at distance r from its centre, the field points along the axis (away from the N pole outside the magnet) and has magnitude $\mu_0 2\mu_m/(4\pi r^3)$.

A permanent magnet is equivalent to a small current loop; its *dipole moment* μ_m (units A m^2) measures its strength. Along the axis of a loop of radius a carrying current I , the Biot–Savart law gives $B = \mu_0 I a^2 / (2(a^2 + z^2)^{3/2})$; far away ($z \gg a$) this becomes $B \approx \mu_0 I a^2 / (2z^3) = \mu_0 (I \pi a^2) / (2\pi z^3)$, and $I \pi a^2$ is the moment μ_m . Hence

$$B(r) = \frac{\mu_0}{4\pi} \frac{2\mu_m}{r^3} \quad (\text{on the axis, } r \gg \text{size of the magnet}). \quad (5)$$

The essential feature is the *inverse cube*: the field of a dipole falls off one power faster than that of a single pole would. (For a real magnet, $\mu_m \approx B_r V / \mu_0$ where B_r is the remanence quoted by the manufacturer and V the volume; a 10 mm $\times$ 3 mm N42 disc has $\mu_m \approx 0.24 \text{ A m}^2$.)

2b. The force on the second magnet



like poles face each other $\Rightarrow \mu_2$ is *anti*-parallel to $B_1 \Rightarrow$ repulsion

Figure 4. Two magnets on a common axis with like poles facing. The field of magnet 1 at magnet 2 points to the right; the moment of magnet 2 points to the left.

A dipole μ_2 in a field B has potential energy $U = -\mu_2 \cdot B$ (lowest when aligned with the field, like a compass needle). With like poles facing, μ_2 is anti-parallel to B_1 , so

$$U(r) = +\mu_m B_1(r) = \frac{\mu_0}{4\pi} \frac{2\mu_m^2}{r^3}. \quad (6)$$

Force is minus the gradient of potential energy:

$$F(r) = -\frac{dU}{dr} = \frac{\mu_0}{4\pi} \frac{6\mu_m^2}{r^4} = \frac{3\mu_0\mu_m^2}{2\pi r^4} \quad (7)$$

The sign is positive, meaning the force *increases* r : repulsion. The physics behind the power: field $\propto r^{-3}$, force $\propto$ its derivative $\propto r^{-4}$.

Common confusion

The r in $F(r)$ is the distance between the *centres* of the two dipoles, not the gap between their faces. That is why the essay defines d centre-to-centre ($d = \text{gap} + \text{thickness}$). And the dipole formula is only exact when r is large compared with the magnet size D ; at $r/D \approx 2$ the real force falls *less* steeply than r^{-4} (essay Section 10.3).

What to say in your own words

(1) Each magnet is modelled as a point dipole. (2) The on-axis field of a dipole is $\propto \mu_m/r^3$. (3) The energy of the second dipole in that field is $+\mu_m B$ because like poles face; differentiating gives a repulsive force $\propto \mu_m^2/r^4$. (4) The model needs $d \gg D$, which is only marginally true here.

3 Step 3: equilibrium and coordinates

Idea in one sentence

With both magnets mounted the repulsion bends each spring outwards by δ until $k\delta = F(d)$; all displacements are then measured from this bent equilibrium, and d is the equilibrium centre-to-centre separation.

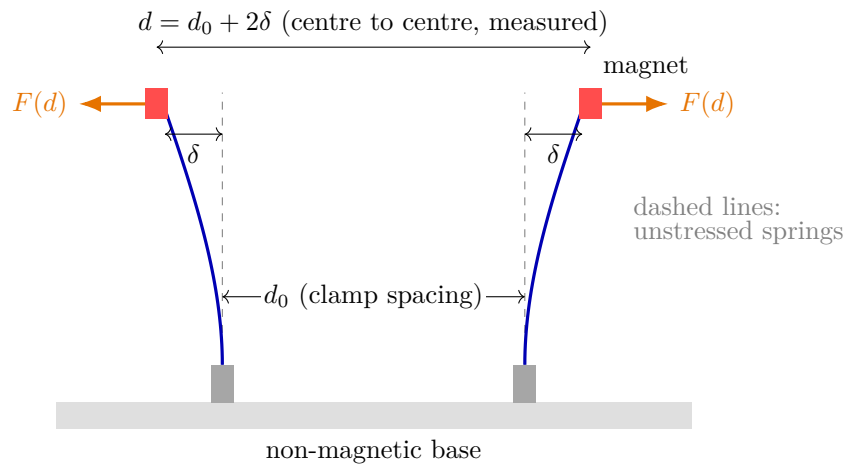


Figure 5. Equilibrium. The repulsion $F(d)$ pushes each tip outwards until the spring's restoring force $k\delta$ balances it.

At rest each spring is bent outwards by δ with $k\delta = F(d)$. If the clamps are d_0 apart, the magnets sit at $d = d_0 + 2\delta$. Finding d from d_0 would mean solving $k(d - d_0)/2 = 3\mu_0\mu_m^2/(2\pi d^4)$ for d —a fifth-degree equation with two unknown constants. The essay avoids this completely by *measuring* d directly at equilibrium (magnets mounted, springs already bent). From then on:

- x_1, x_2 are the displacements of magnets 1 and 2 from their *equilibrium* positions, both counted positive in the same direction (say to the right);
- the instantaneous separation is $r = d + x_2 - x_1$ (if magnet 2 moves right and magnet 1 does not, the gap grows).

Note the unstressed (straight) position of spring 1 is δ to the right of its equilibrium, and that of spring 2 is δ to the left; we will need this for the spring forces in Step 5.

What to say in your own words

Define d as the equilibrium centre-to-centre separation, measured with both magnets in place; explain that this sidesteps the static bending δ (which would otherwise require solving a quintic) and that all displacements are measured from equilibrium.

4 Step 4: linearising the repulsion — the magnetic “spring” γ

Idea in one sentence

For small movements the curved force law can be replaced by its tangent at $r = d$; the slope of that tangent is a stiffness, $\gamma = -F'(d)$: the magnetic repulsion behaves like a compressed spring between the magnets.

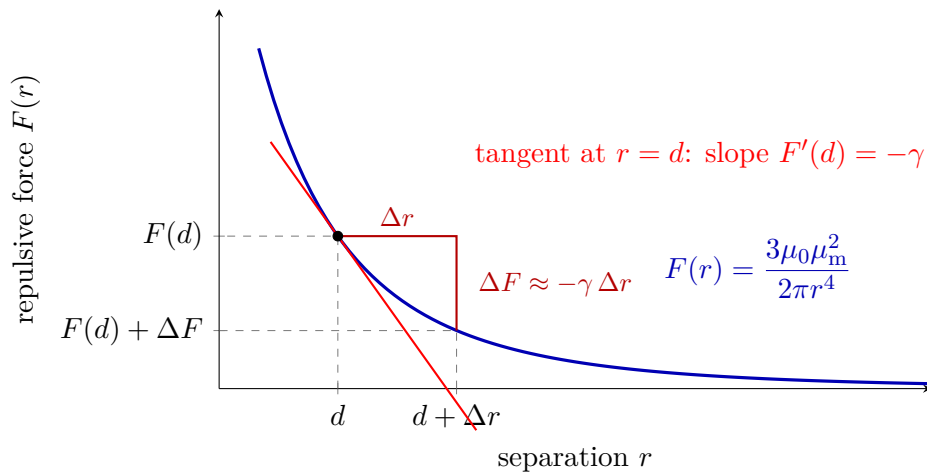


Figure 6. Linearisation. Near the equilibrium separation the force curve is replaced by its tangent. Moving the magnets Δr further apart reduces the repulsion by $\gamma\Delta r$; pushing them closer increases it by the same amount—exactly what a compressed spring of stiffness γ would do.

Take the separation to be $r = d + u$ with u small. A Taylor expansion to first order gives

$$F(d + u) \approx F(d) + F'(d)u, \quad \text{so} \quad F(d + u) - F(d) \approx -\gamma u, \quad \gamma \equiv -F'(d). \quad (8)$$

Differentiating $F \propto r^{-4}$: $\frac{d}{dr}r^{-4} = -4r^{-5}$, hence

$$\boxed{\gamma = -\left.\frac{dF}{dr}\right|_{r=d} = \frac{4F(d)}{d} = \frac{6\mu_0\mu_m^2}{\pi d^5}} \quad (9)$$

γ is called the *magnetic stiffness*: the extra restoring force per unit change of separation. It is positive because a repulsion that *weakens* with distance always pushes the magnets back towards their equilibrium separation—closer means a bigger push apart, further means a smaller push and the springs win. This is the single most important step in the whole theory: it is where the power -5 comes from (one more than the -4 of the force, because γ is a derivative).

Check yourself

What would change if the magnets *attracted*? Then F is negative and grows in magnitude as r shrinks, so $\gamma = -F'(d)$ is *negative*: the magnetic “spring” has negative stiffness and softens the anti-phase mode ($\omega_2 < \omega_1$). If $2|\gamma| > k$ the equilibrium is unstable and the

magnets snap together—which is exactly what happens with attracting magnets on floppy springs.

What to say in your own words

Explain that the displacements are small compared with d , so the force can be replaced by its tangent; that the slope of the tangent, $\gamma = -F'(d) = 4F(d)/d$, is an effective spring constant; and that because $F \propto d^{-4}$, $\gamma \propto d^{-5}$.

5 Step 5: Newton's second law for each magnet

Idea in one sentence

Each magnet feels its own spring pulling it back and the magnetic force from the other magnet; writing $F = ma$ for both, with the linearised magnetic force, gives two coupled linear equations—the same equations as two masses joined by a spring of stiffness γ .

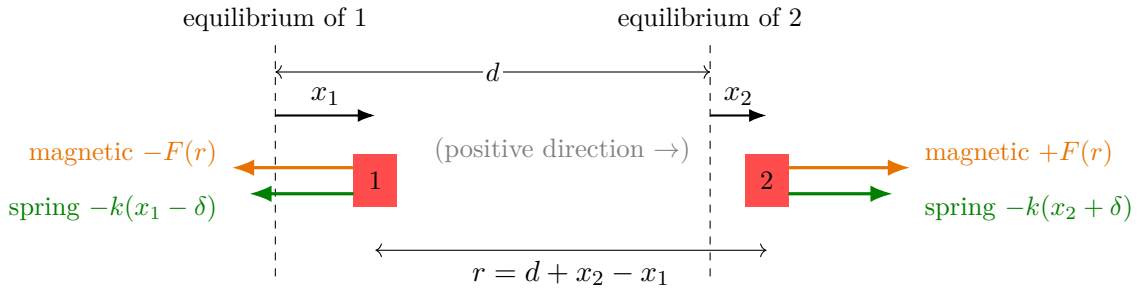


Figure 7. Free-body diagram (horizontal forces only). Both magnets shown displaced to the right. The spring force of each is proportional to the displacement from its *unstressed* position, which is $+\delta$ for spring 1 and $-\delta$ for spring 2 (Figure 5). The magnetic force pushes magnet 1 to the left and magnet 2 to the right, with magnitude $F(r)$ evaluated at the *current* separation.

Magnet 1. Its displacement from the unstressed position is $x_1 - \delta$, so the spring exerts $-k(x_1 - \delta)$; the magnetic force is $-F(d + u)$ with $u = x_2 - x_1$. Newton II:

$$\begin{aligned} m\ddot{x}_1 &= -k(x_1 - \delta) - F(d + u) \\ &= -kx_1 + \underbrace{k\delta}_{=F(d)} - F(d + u) = -kx_1 + [F(d) - F(d + u)] \approx -kx_1 + \gamma u. \end{aligned} \quad (10)$$

The equilibrium condition $k\delta = F(d)$ has cancelled the constant part of the magnetic force; only the *change* of the force, $-\gamma u$ from Step 4, survives. **Magnet 2** is the mirror image: spring force $-k(x_2 + \delta)$, magnetic force $+F(d + u)$, giving $m\ddot{x}_2 = -kx_2 - \gamma u$. Together:

$$m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1) \quad (11)$$

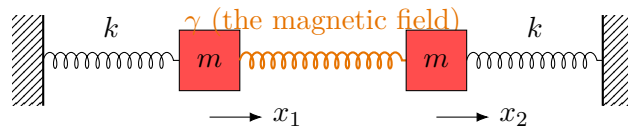


Figure 8. Mechanical equivalent of equations (11): two equal masses on equal springs k , joined by a third spring of stiffness γ . The middle spring exerts $\gamma(x_2 - x_1)$ on the left mass and the opposite on the right mass—exactly the coupling terms.

Common confusion

Students often write the magnetic force as $-\gamma x_1$. It is not: the magnetic force depends on the *separation* $x_2 - x_1$, not on where magnet 1 is. If both magnets move together ($x_1 = x_2$) the separation does not change and the magnetic force does not change at all.

What to say in your own words

Draw the free-body diagram, state the two forces on each magnet, apply Newton II, use $k\delta = F(d)$ to remove the constant force, and use Step 4 to linearise. Point out that the result is identical to two masses coupled by a spring of stiffness γ .

6 Step 6: normal modes

Idea in one sentence

The two equations are decoupled by adding and subtracting them: the sum $x_1 + x_2$ and the difference $x_1 - x_2$ each obey a simple harmonic equation with its own frequency. These are the two normal modes.

Add the two equations of (11): the coupling terms cancel,

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2) \implies m\ddot{\eta} = -k\eta, \quad \eta \equiv x_1 + x_2. \quad (12)$$

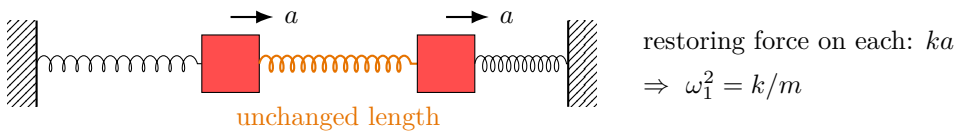
Subtract them: the coupling terms add,

$$m(\ddot{x}_1 - \ddot{x}_2) = -k(x_1 - x_2) + 2\gamma(x_2 - x_1) = -(k + 2\gamma)(x_1 - x_2) \implies m\ddot{\xi} = -(k + 2\gamma)\xi, \quad \xi \equiv x_1 - x_2. \quad (13)$$

Each of η and ξ is a simple harmonic oscillator ($\ddot{q} = -\omega^2 q$), so

$$\omega_1^2 = \frac{k}{m} \quad (\text{in-phase, } \eta), \quad \omega_2^2 = \frac{k + 2\gamma}{m} \quad (\text{anti-phase, } \xi) \quad (14)$$

In-phase mode ($x_1 = x_2$, $\xi = 0$): both move together



Anti-phase mode ($x_1 = -x_2$, $\eta = 0$): mirror images

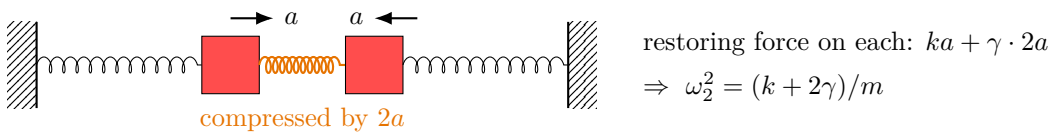


Figure 9. The two normal modes and the origin of the factor 2. In the in-phase mode the coupling spring keeps its length, so the magnetic force never changes and each mass feels only its own spring. In the anti-phase mode a displacement a of each mass changes the separation by $2a$, so the coupling spring pushes each mass with $2\gamma a$ in addition to ka .

Why “modes”? If the system starts exactly in one of these patterns it stays in it forever, oscillating at one pure frequency: in-phase at $f_1 = \omega_1/2\pi$, anti-phase at $f_2 = \omega_2/2\pi$. Any other

motion is a mixture of the two (Step 8). Note that f_1 is exactly the frequency of a single spring on its own, $\sqrt{k/m}/2\pi$: in the in-phase mode the magnets might as well not be there. And $f_2 > f_1$ because the anti-phase mode has the extra stiffness 2γ .

Check yourself

Two limits. As $d \rightarrow \infty$, $\gamma \rightarrow 0$ and both frequencies become $\sqrt{k/m}$: uncoupled springs. As d shrinks, γ grows as d^{-5} , f_2 rises steeply while f_1 does not move. With your data at 21 mm, $2\gamma/k = (f_2/f_1)^2 - 1 = (1.063/0.823)^2 - 1 = 0.67$, i.e. the magnetic spring is about one third as stiff as a leaf spring; at 40 mm it is 0.02.

What to say in your own words

Define the normal coordinates $\eta = x_1 + x_2$ and $\xi = x_1 - x_2$; show that adding/subtracting the equations of motion gives two *independent* SHM equations; state the two frequencies; explain physically why the in-phase frequency does not contain γ and why the anti-phase one contains 2γ (separation changes twice as fast as each magnet moves).

7 Step 7: the relation that is tested

Idea in one sentence

Subtracting the squared mode frequencies removes k entirely and leaves γ/m , which the dipole model says is proportional to d^{-5} ; on log-log axes this is a straight line of gradient -5 .

With $\omega = 2\pi f$, equations (14) give

$$f_2^2 - f_1^2 = \frac{\omega_2^2 - \omega_1^2}{4\pi^2} = \frac{2\gamma}{4\pi^2 m} = \frac{\gamma}{2\pi^2 m} = \frac{3\mu_0\mu_m^2}{\pi^3 m} d^{-5} \equiv C d^{-5} \quad (15)$$

(the last step inserts $\gamma = 6\mu_0\mu_m^2/(\pi d^5)$ from Step 4). Taking natural logarithms,

$$\ln(f_2^2 - f_1^2) = \ln C - 5 \ln d. \quad (16)$$

So a graph of $\ln(f_2^2 - f_1^2)$ against $\ln d$ should be a straight line with gradient -5 and intercept $\ln C$.

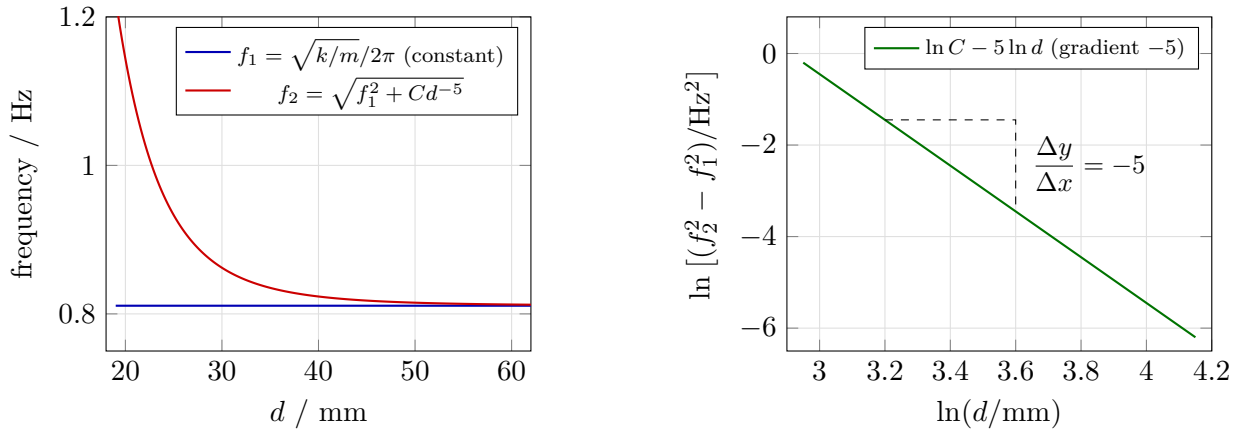


Figure 10. What the model predicts (drawn with $f_1 = 0.811$ Hz and the C fitted to your data). Left: f_1 is flat, f_2 falls towards it as $d^{-5/2}$ -ish and merges with it beyond ~ 45 mm. Right: the coupling term on log-log axes is a straight line of gradient exactly -5 .

Why subtract the squares, rather than just look at f_2 ? Three reasons. (i) k cancels: you do not need to know the spring constant, Young’s modulus, or the strip dimensions to test the force law. (ii) f_1 is measured in the *same* record as f_2 , so any slow drift of the springs (temperature, clamp creep) affects both and largely cancels. (iii) The result is *linear* in γ , so the exponent of d appears directly as the gradient of the ln–ln plot. The gradient tests the *form* of the force law ($r^{-4} \Rightarrow -5$); the intercept $\ln C = \ln(3\mu_0\mu_m^2/\pi^3m)$ tests its *magnitude* and needs μ_m and m .

Units check. $[\mu_0\mu_m^2] = \text{T m A}^{-1} \cdot \text{A}^2\text{m}^4 = \text{T A m}^5$, and $1 \text{ T} = 1 \text{ kg A}^{-1}\text{s}^{-2}$, so $[\mu_0\mu_m^2] = \text{kg m}^5\text{s}^{-2}$. Dividing by m (kg) and d^5 (m^5) leaves s^{-2} : the units of f^2 . C itself is in m^5s^{-2} .

Check yourself

Put numbers in. At $d = 21 \text{ mm}$: $f_2^2 - f_1^2 = 1.063^2 - 0.823^2 = 0.4526 \text{ Hz}^2$, so $\gamma/m = 2\pi^2 \times 0.4526 = 8.9 \text{ s}^{-2}$, while $k/m = (2\pi \times 0.823)^2 = 26.7 \text{ s}^{-2}$. If the dipole law held exactly, going from 21 to 40 mm should divide the coupling term by $(40/21)^5 = 25$, to 0.018 Hz^2 ; the measured value is 0.0145 Hz^2 .

What to say in your own words

Show the subtraction, emphasise that k drops out, write $f_2^2 - f_1^2 = Cd^{-5}$ with C explicit, take logs, and say what the gradient and the intercept each test.

8 Step 8: what you actually see — beats

Idea in one sentence

After release the motion is a *mixture* of the two modes. The frequencies in the mixture (f_1, f_2) are fixed by the system; the *amounts* of each mode are fixed by how you start it. An FFT peak height is simply the amount of that mode in the record, and the sum of two cosines of nearby frequency is a beat.

8a. The general motion has two “amounts”, η_0 and ξ_0

Step 6 showed that $\eta = x_1 + x_2$ and $\xi = x_1 - x_2$ each obey their own SHM equation, so the most general motion is

$$\eta(t) = \eta_0 \cos(\omega_1 t + \phi_1), \quad \xi(t) = \xi_0 \cos(\omega_2 t + \phi_2),$$

with four constants fixed by the four initial conditions (two positions, two velocities). Both springs start from rest, $\dot{\eta}(0) = \dot{\xi}(0) = 0$, which forces $\phi_1 = \phi_2 = 0$. The two *mode amplitudes* are then just the initial values of the normal coordinates:

$$\eta_0 = x_1(0) + x_2(0), \quad \xi_0 = x_1(0) - x_2(0). \quad (17)$$

Transforming back with $x_1 = (\eta + \xi)/2$ and $x_2 = (\eta - \xi)/2$:

$$x_1(t) = \frac{\eta_0}{2} \cos \omega_1 t + \frac{\xi_0}{2} \cos \omega_2 t, \quad x_2(t) = \frac{\eta_0}{2} \cos \omega_1 t - \frac{\xi_0}{2} \cos \omega_2 t \quad (18)$$

Each spring’s displacement is *two cosines*, one at each mode frequency, with amplitudes $\eta_0/2$ and $\xi_0/2$. Different ways of starting give different amounts:

how you start it	$x_1(0), x_2(0)$	η_0, ξ_0	what you see
both pushed the same way	A, A	$2A, 0$	pure in-phase mode: one frequency f_1 , no beats
pushed opposite ways	$A, -A$	$0, 2A$	pure anti-phase mode: one frequency f_2 , no beats
one displaced, other exactly at rest	$A, 0$	A, A	both modes equally: beats that go down to zero
one <i>held</i> , other has drifted	A, x_{20}	$A + x_{20}, A - x_{20}$	both modes unequally: beats that do not reach zero

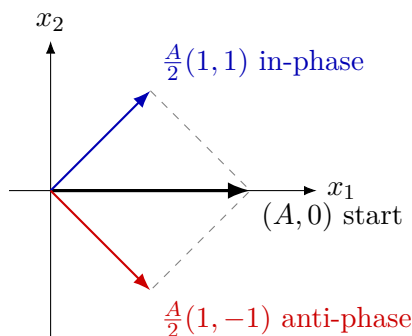


Figure 11. The third row of the table as a picture: the starting configuration $(x_1, x_2) = (A, 0)$ is the sum of equal amounts of the in-phase pattern $(1, 1)$ and the anti-phase pattern $(1, -1)$. Each pattern then oscillates at its own frequency. For the fourth row the black arrow tilts up to (A, x_{20}) and the blue component becomes longer than the red one.

8b. What the height of an FFT peak means

A Fourier transform takes a record and works out which cosines it is made of; for each frequency it reports the *amplitude* of the cosine at that frequency. Equation (18) says that $x_1(t)$ is made of exactly two cosines, so its spectrum has exactly two peaks, and (for a record long enough to separate them, $T \gg 1/(f_2 - f_1)$) their heights are proportional to the two amplitudes:

$$\frac{\text{height of the } f_2 \text{ peak}}{\text{height of the } f_1 \text{ peak}} = \frac{\xi_0/2}{\eta_0/2} = \frac{\xi_0}{\eta_0}. \quad (19)$$

That is the *entire* content of the “equal heights” remark: the peaks are equal if and only if $\eta_0 = \xi_0$, i.e. if and only if $x_2(0) = 0$. It is a statement about the starting condition, not about the modes. Two consequences matter for the essay:

- The *positions* of the peaks, f_1 and f_2 , do not depend on the starting condition at all. That is why one FFT of one spring gives both mode frequencies whatever the exact release—the method never needed equal heights, only that both modes are present.
- The *heights* tell you about the starting condition and (8f) about whether the two springs are identical.

8c. Why your peaks are *not* equal — and why that is correct

You observed that the f_2 peak is only about 0.6 of the f_1 peak and that the beats do not go down to zero. Both follow from how a spring is really released. You do not flick spring 1 instantaneously from 0 to A : you *hold* it at A for a moment and then let go. While it is held the separation of the magnets is different, so the repulsion on spring 2 is different, and spring 2 quietly moves to a new rest position *before* you release. Setting $\ddot{x}_2 = 0$ in the equation of motion of spring 2 (Step 5) with $x_1 = A$:

$$0 = -kx_2 - \gamma(x_2 - A) \quad \implies \quad x_{20} = \frac{\gamma}{k + \gamma} A,$$

in the same direction as A (spring 2 is pushed away if you push spring 1 towards it, and follows if you pull spring 1 away). By (17), $\eta_0 = A + x_{20}$ and $\xi_0 = A - x_{20}$, so by (19)

$$\frac{\text{height at } f_2}{\text{height at } f_1} = \frac{A - x_{20}}{A + x_{20}} = \frac{k}{k + 2\gamma} = \frac{\omega_1^2}{\omega_2^2} = \left(\frac{f_1}{f_2}\right)^2 \quad (20)$$

With your $f_1 = 0.802$ and $f_2 = 0.996$ Hz this is 0.65; you measured about 0.58. The remaining difference is about the size of the non-linear correction: with the full r^{-4} force instead of its tangent, a 10 mm hold gives 0.49 if spring 1 was pushed *towards* spring 2 and 0.71 if it was pulled *away* ($A = 10$ mm is not small compared with $d = 23$ mm). So a ratio around 0.6 is exactly what a real hold-and-release should give. It is not an error to be fixed, it does not move f_1 or f_2 , and in the essay you can quote (20) as a small extra check of the model. The envelope minima follow too: the amplitude swings between $(\eta_0 + \xi_0)/2 = A$ and $(\eta_0 - \xi_0)/2 = x_{20}$, so minimum/maximum = $x_{20}/A = \gamma/(k + \gamma) \approx 0.2$ instead of 0.

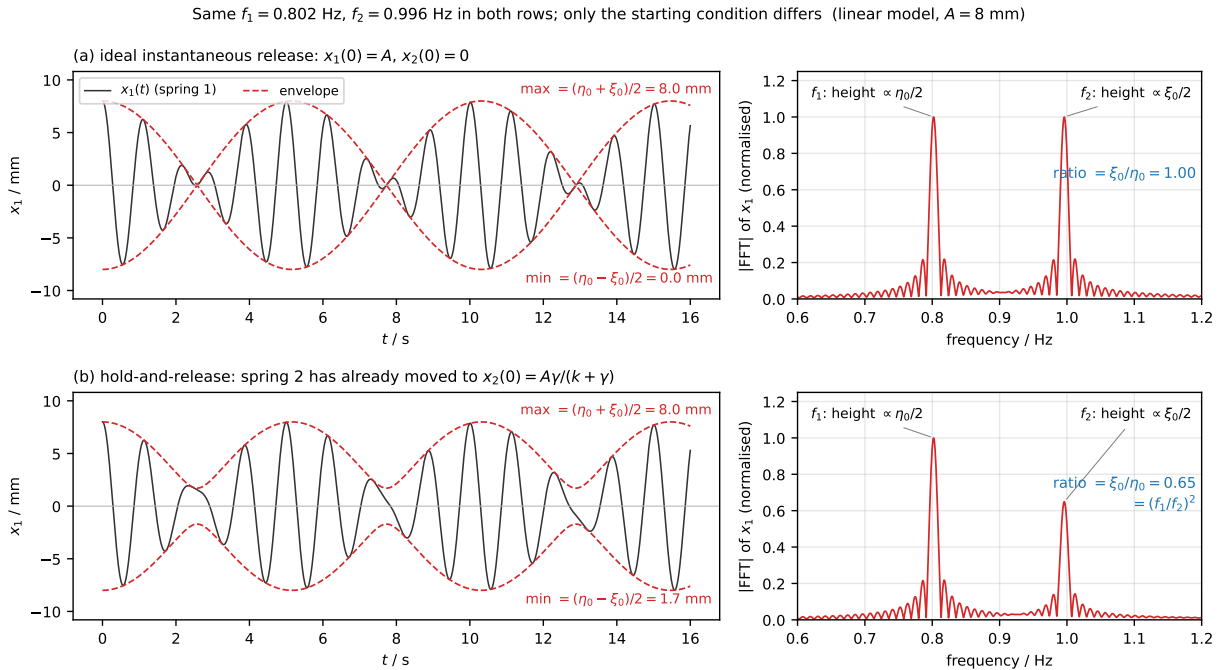


Figure 12. The same system ($f_1 = 0.802$, $f_2 = 0.996$ Hz) started in two ways (linear model). Top: idealised instantaneous release, $x_2(0) = 0$: equal mode amplitudes, equal FFT peaks, beats that touch zero. Bottom: hold-and-release, $x_2(0) = A\gamma/(k + \gamma) = 1.7$ mm: peak ratio $(f_1/f_2)^2 = 0.65$ and envelope minima of 1.7 mm instead of zero. The peak *positions* are identical in both rows.

8d. Two cosines make a beat: the rotating-arrow picture

Write each cosine as the horizontal projection of a rotating arrow (a phasor). Arrow 1 has length $\eta_0/2$ and turns at f_1 revolutions per second; arrow 2 has length $\xi_0/2$ and turns at f_2 revolutions per second; x_1 is the horizontal projection of their vector sum (Figure 13). Because arrow 2 turns slightly faster, the angle between them grows steadily, by $f_2 - f_1$ revolutions every second. When they point the same way the sum is long, $(\eta_0 + \xi_0)/2$; half a relative turn later they oppose each other and the sum is short, $|\eta_0 - \xi_0|/2$; after one full relative turn they are aligned again. So the amplitude of the fast oscillation swells and shrinks with period

$$T_{\text{beat}} = \frac{1}{f_2 - f_1}, \quad f_{\text{beat}} = f_2 - f_1 \quad (21)$$

whatever the two lengths are; the lengths only decide how deep the minima go (zero for equal lengths, x_{20} for the hold-and-release). Meanwhile the sum-arrow itself turns at about the average rate, which is the fast oscillation at $f_{\text{avg}} = (f_1 + f_2)/2$ (exactly that for equal lengths).

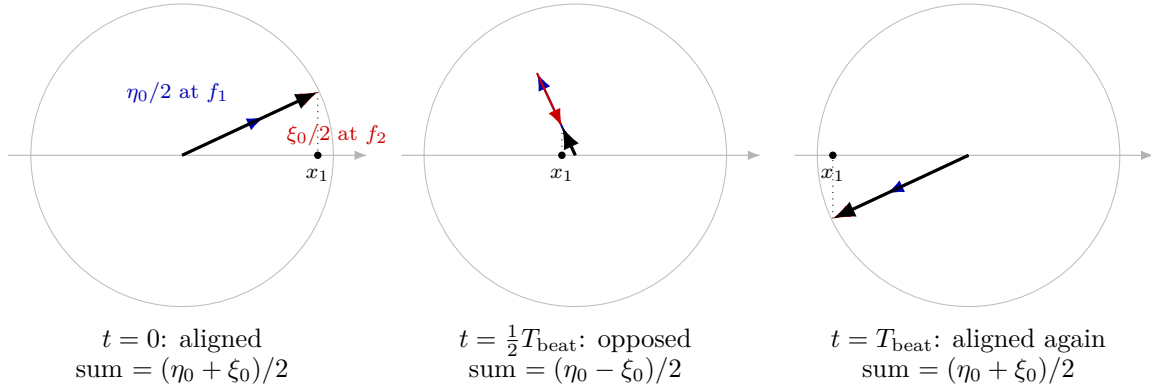


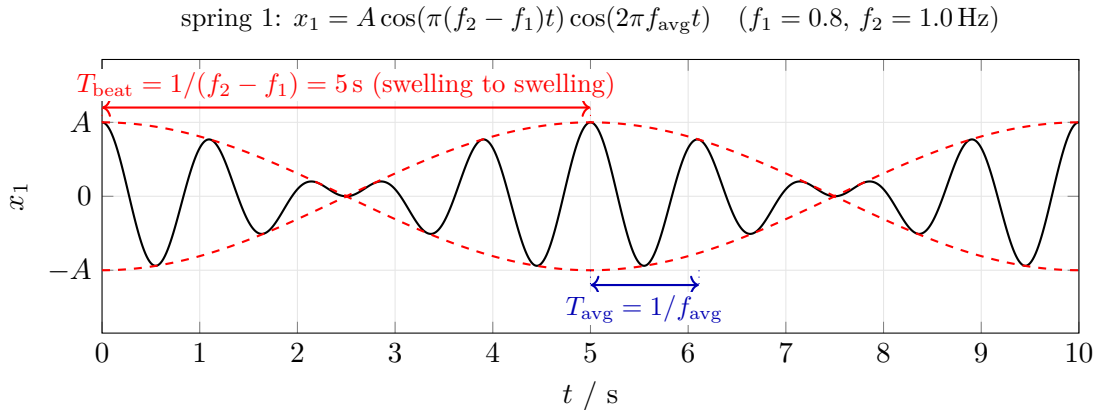
Figure 13. Beats as rotating arrows. Blue: length $\eta_0/2$, turning at f_1 rev/s. Red (drawn head-to-tail): length $\xi_0/2$, turning at f_2 rev/s. Black: their sum; x_1 is its horizontal projection. The relative angle grows by one full turn every $1/(f_2 - f_1)$ seconds, so the length of the black arrow—the amplitude of the fast oscillation—repeats with exactly that period. Lengths drawn for $\xi_0/\eta_0 = 0.65$ (the hold-and-release); the sum never shrinks to zero.

8e. The same thing with a trig identity (equal amplitudes)

For $\eta_0 = \xi_0 = A$ the product-to-sum identity $\cos \alpha + \cos \beta = 2 \cos \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$, with $\alpha = 2\pi f_1 t$ and $\beta = 2\pi f_2 t$, turns x_1 into

$$x_1(t) = A \cos(\pi(f_2 - f_1)t) \cos(2\pi f_{\text{avg}}t), \quad f_{\text{avg}} = \frac{f_1 + f_2}{2} \quad (22)$$

Read this as *[slowly varying amplitude] × [fast oscillation]*: the second factor is the fast oscillation at the average frequency, the first is the envelope (Figure 14).



spring 2: $x_2 = A \sin(\pi(f_2 - f_1)t) \sin(2\pi f_{\text{avg}}t)$ — full amplitude exactly when spring 1 is at rest

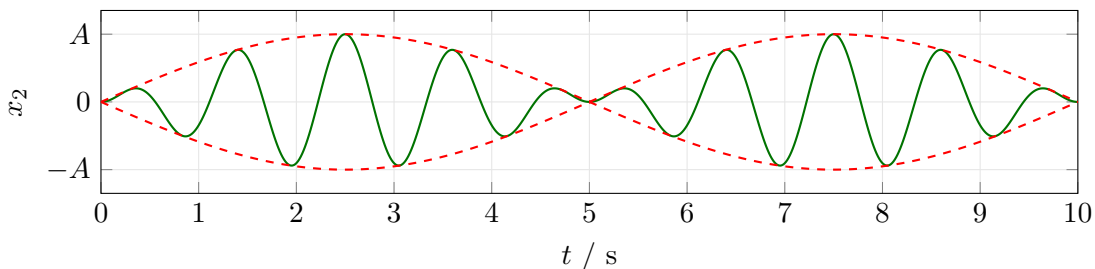


Figure 14. Beats for equal mode amplitudes. Top: the released spring (black) and its envelope (red dashed). The envelope cosine has frequency $(f_2 - f_1)/2$, but its *modulus* peaks twice per cycle, so the swellings recur every $T_{\text{beat}} = 1/(f_2 - f_1)$ —the same conclusion as the arrow picture. Bottom: spring 2 is a quarter of a beat behind; it has all the energy when spring 1 has none.

Common confusion

“Is the beat frequency $f_2 - f_1$ or $(f_2 - f_1)/2$?” Both numbers appear and they mean different things. The *envelope function* $\cos(\pi(f_2 - f_1)t)$ has frequency $(f_2 - f_1)/2$. But an amplitude cannot be negative: what you see is $|\cos|$, which has maxima twice per cycle. The time between two swellings—what you measure on the Tracker plot—is $1/(f_2 - f_1)$ (the arrow picture gives this directly, with no sign ambiguity). So the observable beat frequency is

$$f_{\text{beat}} = f_2 - f_1 = \frac{1}{T_{\text{beat}}}, \quad f_2^2 - f_1^2 = (f_2 - f_1)(f_2 + f_1) = 2 f_{\text{beat}} f_{\text{avg}}.$$

Use this convention consistently; it is the one used in the essay (Figure 3, Section 5.4, Graph 4).

8f. Spring 2, and a test for identical springs

Spring 2 contains the same two cosines with the sign of the anti-phase term reversed (equation (18)). Three consequences:

1. Its FFT has peaks at the *same* f_1, f_2 , with the *same* height ratio ξ_0/η_0 —for identical springs this holds for any starting condition, because each mode has the same amplitude in both springs (mode shapes $(1, 1)$ and $(1, -1)$).
2. For equal amplitudes, $x_2 = A \sin(\pi(f_2 - f_1)t) \sin(2\pi f_{\text{avg}}t)$: its envelope is a quarter of a beat behind that of spring 1, so spring 2 is at full swing when spring 1 is quiet (Figure 14, bottom). This is the energy transfer of the introduction.
3. If the springs are *not* identical, the mode shapes tilt to (a_1, b_1) and $(-b_1, a_1)$ with $a_1 \neq b_1$, and the height ratios of the two springs differ: $r_2/r_1 = (a_1/b_1)^2$, with $a_1/b_1 - b_1/a_1 = 2\Delta/\kappa$ in the notation of essay Section 10.4. Your spectra show r_2 slightly larger than r_1 (about 10%), which corresponds to a stiffness mismatch of order 1% with spring 2 the stiffer one—the cleanest test of assumption (c) you have, and much smaller than the 7% the detuned fit suggested.

8g. Predictions for f_{avg} and f_{beat}

Combining $f_2 = \sqrt{f_1^2 + Cd^{-5}}$ with the definitions:

$$f_{\text{beat}} = \sqrt{f_1^2 + Cd^{-5}} - f_1 = \frac{Cd^{-5}}{2f_{\text{avg}}}, \quad f_{\text{avg}} = \frac{f_1 + \sqrt{f_1^2 + Cd^{-5}}}{2}. \quad (23)$$

Because f_{avg} only varies between f_1 and about $1.15f_1$, the beat frequency falls almost exactly as d^{-5} (for weak coupling $f_{\text{beat}} \approx Cd^{-5}/2f_1$), while the average frequency is nearly constant and rises only at small d . With your numbers: $T_{\text{beat}} \approx 4\text{ s}$ at 21 mm, $\approx 110\text{ s}$ at 40 mm, and the dipole law predicts about ten *minutes* at 60 mm—which is why no beats (and no second peak) were seen there in a 30 s record.

What to say in your own words

- (1) After release the motion is a superposition of the two modes; starting from rest, the mode amplitudes are $\eta_0 = x_1(0) + x_2(0)$ and $\xi_0 = x_1(0) - x_2(0)$. (2) Each spring’s displacement is therefore two cosines at f_1 and f_2 with amplitudes $\eta_0/2$ and $\xi_0/2$, so an FFT of one spring

shows both frequencies; the peak heights are in the ratio ξ_0/η_0 and say nothing about f_1, f_2 themselves. (3) For the hold-and-release actually used, $x_2(0) = A\gamma/(k + \gamma)$, giving a height ratio $(f_1/f_2)^2 \approx 0.65$ and beats whose minima are not zero—consistent with what was observed. (4) Two cosines of nearby frequency beat: fast oscillation at $f_{\text{avg}} = (f_1 + f_2)/2$, amplitude swelling every $T_{\text{beat}} = 1/(f_2 - f_1)$, so $f_{\text{beat}} = f_2 - f_1$ and $f_2^2 - f_1^2 = 2f_{\text{beat}}f_{\text{avg}}$. (5) Spring 2 carries the same two frequencies with the anti-phase term reversed, so it swings most when spring 1 is quiet; identical springs give the same height ratio in both spectra, non-identical springs do not.

9 The assumptions, and why each one matters

Assumption	Where it entered	What happens if it fails
(a) magnets are point dipoles ($d \gg D$)	Step 2, $F \propto r^{-4}$	force falls less steeply at small d ; gradient shallower than -5 (essay 10.3)
(b) displacements small ($A \ll d$)	Step 4, tangent instead of curve	frequency depends on amplitude; f_2 pushed up at small d (essay 10.2)
(c) the two springs are identical	Step 6, adding/subtracting works only if k, m equal	f_1 no longer constant; peak-height ratio differs between the two springs; coupling term saturates at large d (essay 10.4)
(d) one spring = one lumped mass	Step 1	small error in m ; higher bending modes appear as extra (much higher) peaks
(e) no damping	Steps 6, 8	amplitude decays; FFT peaks broaden; frequencies shift negligibly
(f) magnets stay coaxial	Step 2 (on-axis field)	tip of a bent cantilever tilts; $\lesssim 1\%$ effect, independent of d
(g) the FFT returns the true frequencies	not in the theory, but in the test of it	two close peaks pull together/merge for short records (essay 6, 10.1)

10 Symbols

Symbol	Meaning	Symbol	Meaning
L, b, h	length, width, thickness of a spring	d	equilibrium centre-to-centre separation
$E, I = bh^3/12$	Young's modulus, second moment of area	δ	static outward bend of each spring, $k\delta = F(d)$
$k = 3EI/L^3$	tip stiffness of a spring	x_1, x_2	displacements from equilibrium (same sign convention)
M, μ	magnet mass, strip mass per length	$r = d + x_2 - x_1$	instantaneous separation
$m = M + \frac{33}{140}\mu L$	effective oscillating mass	$\gamma = -F'(d)$	magnetic stiffness $= 6\mu_0\mu_m^2/\pi d^5$
μ_m	dipole moment of a magnet	$\eta = x_1 + x_2, \xi = x_1 - x_2$	normal coordinates
μ_0	permeability of free space	$\omega_1^2 = k/m, \omega_2^2 = (k + 2\gamma)/m$	mode angular frequencies
$F(r) = 3\mu_0\mu_m^2/2\pi r^4$	repulsive force	$C = 3\mu_0\mu_m^2/\pi^3 m$	constant in $f_2^2 - f_1^2 = Cd^{-5}$
$f_{\text{avg}} = (f_1 + f_2)/2$	average frequency	$f_{\text{beat}} = f_2 - f_1 = 1/T_{\text{beat}}$	beat frequency

11 The argument in ten sentences (skeleton to expand in your own words)

1. A clamped leaf spring pushed at its tip obeys Hooke's law with $k = 3EI/L^3$, and the tip mass that oscillates is the magnet plus 33/140 of the strip's mass.
2. Each spring alone is therefore a simple harmonic oscillator with $\omega_0^2 = k/m$.
3. Treating each magnet as a point dipole, the on-axis field falls as r^{-3} and the repulsion between the two like-facing magnets is $F = 3\mu_0\mu_m^2/(2\pi r^4)$.
4. At equilibrium the repulsion bends each spring outwards by δ with $k\delta = F(d)$; d is defined and measured in this state, and displacements x_1, x_2 are counted from it.
5. For small displacements the change in the repulsion is proportional to the change in separation, $\Delta F = -\gamma\Delta r$, with $\gamma = -F'(d) = 4F(d)/d \propto d^{-5}$: the magnetic field acts as a spring of stiffness γ between the magnets.
6. Newton's second law for each magnet, after the equilibrium force cancels, gives $m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1)$ and $m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1)$ —two masses joined by a third spring.
7. Adding and subtracting decouples them: $x_1 + x_2$ oscillates at $\omega_1^2 = k/m$ (in phase, magnets keep their distance, coupling irrelevant) and $x_1 - x_2$ at $\omega_2^2 = (k + 2\gamma)/m$ (anti-phase, separation changes twice as fast as each magnet moves, hence 2γ).
8. Subtracting the squared frequencies eliminates k : $f_2^2 - f_1^2 = \gamma/(2\pi^2 m) = Cd^{-5}$, so $\ln(f_2^2 - f_1^2)$ against $\ln d$ should be a straight line of gradient -5 ; the intercept gives $C = 3\mu_0\mu_m^2/\pi^3 m$.
9. Releasing one spring from rest excites both modes (with amplitudes $\eta_0 = x_1(0) + x_2(0)$ and $\xi_0 = x_1(0) - x_2(0)$, equal only for an instantaneous release), so each magnet's motion is $\frac{\eta_0}{2} \cos \omega_1 t \pm \frac{\xi_0}{2} \cos \omega_2 t$: a fast oscillation at $f_{\text{avg}} = (f_1 + f_2)/2$ whose amplitude beats at $f_{\text{beat}} = f_2 - f_1$, with energy passing back and forth between the springs; an FFT of either spring shows two peaks at f_1 and f_2 whose height ratio ξ_0/η_0 reflects only the starting condition.
10. The derivation assumed point dipoles, small amplitudes, identical springs, a lumped mass, no damping and coaxial magnets; each assumption is examined in the evaluation.

Answers to the “check yourself” questions and two more

Why does k cancel? Because ω_1^2 and ω_2^2 both contain k/m and differ only by $2\gamma/m$. This is the reason for using $f_2^2 - f_1^2$ rather than f_2 alone.

Why the factor 2 in $k + 2\gamma$? In the anti-phase mode each magnet moves a but the separation changes by $2a$; the magnetic spring responds to the separation, so it exerts $2\gamma a$ on each magnet.

Why does f_1 not depend on d ? In the in-phase mode the separation never changes, so the magnetic force stays at its constant equilibrium value, which is already balanced by the static bend δ .

Why -5 and not -4 ? The force is $\propto r^{-4}$, but the frequency shift depends on the *stiffness*, the derivative of the force, which is $\propto r^{-5}$.

Why are the two FFT peaks not equal in height? Their heights are the amplitudes $\eta_0/2$ and $\xi_0/2$ of the two modes, fixed by the starting positions: equal only if $x_2(0) = 0$. Holding spring 1 at A lets spring 2 drift to $x_{20} = A\gamma/(k + \gamma)$ first, so the ratio is $(A - x_{20})/(A + x_{20}) = (f_1/f_2)^2 \approx 0.65$, and the beats no longer reach zero. Neither effect changes f_1 or f_2 .